Functions

Tuesday, September 29, 2026 6:40 PM

Straight Line

$y = mx + c$ / $y - y_1 = m(x - x_1)$

m - gradient

c - y-intercept

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m > 0$, the graph is increasing

$m < 0$, the graph is decreasing

$m = 5 > 0$

$m = -6 < 0$

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

Parabola

$y = f(x) = a(x \pm p)^2 \pm q = ax^2 + bx + c$

↳ y-intercept

TP $(p; q)$

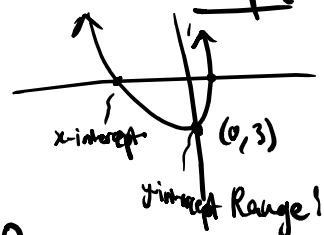
↳ axis of symmetry.

TP: $x = -\frac{b}{2a}$; $y = f(-\frac{b}{2a})$

$a < 0$: concave down ∩

$a > 0$: concave up ∪

Example



Range: $y \in [3, \infty)$

Domain: $x \in \mathbb{R}$

Range: $y \in [q; \infty)$ / $y \geq q$ if $a > 0$

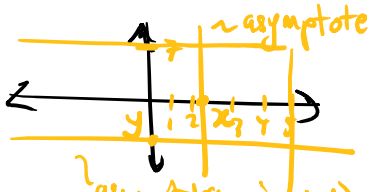
$y \in (-\infty, q]$ / $y \leq q$ if $a < 0$

Hyperbola

$y = \frac{a}{x \pm p} \pm q$ ← $x + 2 = 0$, $x = -2$

Vertical asymptote: $x = p$

Horizontal asymptote: $y = q$



$y = \frac{a}{x}$

~ Hyperbola

example

$f(x) = \frac{2}{x-5} + 7$

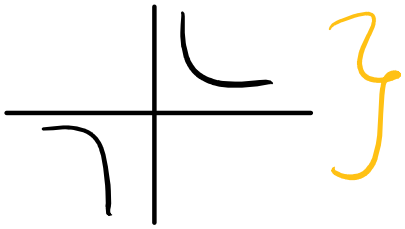
Vertical asymptote: $x - 5 = 0$, $x = 5$

Horizontal asymptote: $y = 7$

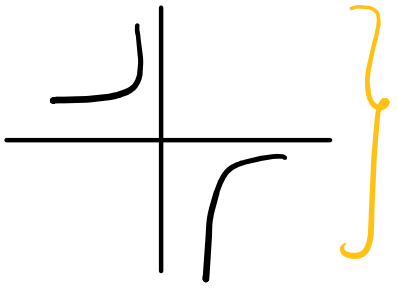


Horizontal asymptote: $y=q$

$a>0$:



$a<0$:



Example: Domain: $x \in \mathbb{R}, x \neq 5$

Example, Range: $y \in \mathbb{R}, y \neq 7$

Axis of symmetry

$\begin{cases} y = x+c \\ y = -x+c \end{cases}$

Domain: $x \in \mathbb{R}, x \neq \pm p$

Range: $y \in \mathbb{R}, y \neq q$

$y = f(x)$

Exponential

$1^{x+5} = 1$

$y = a \cdot b^{x \pm p} \pm q$

$b > 0$ and $b \neq 1$

Horizontal asymptote: $y = q$

$a>0$: graph is above the asymptote.

$a<0$: graph is below the asymptote.

Domain: $x \in \mathbb{R}$

Range: $y \in (q; \infty) / y > q$ if $a>0$

$y \in (-\infty, q) / y < q$ if $a<0$

Examples

1. A straight-line graph passes through two points, namely $(-2;-3)$ and $(5;7)$.

(a) Determine the equation of the graph.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-(-3)}{5-(-2)} = \frac{10}{7}$$

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

$$y - y_1 = m(x - x_1)$$
$$y-(-3) = \frac{10}{7}(x-(-2))$$
$$y+3 = \frac{10}{7}(x+2)$$
$$y = \frac{10}{7}x + \frac{20}{7} - 3$$
$$y = \frac{10}{7}x - \frac{1}{7}$$

x-intercept: $y = 0$

$$\frac{10}{7}x - \frac{1}{7} = 0$$
$$\left[\frac{10}{7}x = \frac{1}{7}\right] \times \frac{7}{10}$$
$$x = \frac{1}{10}$$

2. Given: $f(x) = -2x^2 + x + 6$

Domain: $x \in \mathbb{R}$

Range: $y \in (-\infty, \frac{49}{8})$

(a) Determine the y-intercept. $y = 6$

(b) Determine the x-intercepts. $y = 0$

$$-2x^2 + x + 6 = 0$$
$$(-2x-3)(x-2) = 0$$
$$x = -\frac{3}{2} \text{ or } x = 2$$

(c) Determine the coordinates of the TP. $x = \frac{-b}{2a} = \frac{-1}{2(-2)} = \frac{1}{4}$

$$f\left(\frac{1}{4}\right) = -2\left(\frac{1}{4}\right)^2 + \frac{1}{4} + 6 = \frac{49}{8}$$

$$TP\left(\frac{1}{4}; \frac{49}{8}\right)$$

$IP\left(\frac{1}{4}; \frac{17}{8}\right)$

3. Given the equation of the hyperbola: $f(x) = \frac{-4}{2-x} - 1$

Domain: $x \in \mathbb{R}, x \neq 2$

Range: $y \in \mathbb{R}, y \neq -1$

(a) Write down the equations of the asymptotes of f.

Vertical asymptote: $2 - x = 0$

$x = 2$

Horizontal asymptote: $y = -1$

(b) Determine the intercepts of the graph of f with the axis.

y-intercept: $y = f(0) = \frac{-4}{2-0} - 1 = -3$

x-intercept: $0 = \frac{-4}{2-x} - 1$

$1 = \frac{-4}{2-x}$

$2 - x = -4$

$-x = -6$

$x = 6$

4. Given the equation of the graph: $f(x) = \frac{x+6}{x+2}$

(a) Express $f(x)$ in the form of $f(x) = \frac{a}{x+p} + q$

$f(x) = \frac{x+6}{x+2} = \frac{x+2+4}{x+2} = \frac{x+2}{x+2}\frac{1}{} + \frac{4}{x+2} = \frac{4}{x+2} + 1$

Vertical asymptote: $x = -2$

Horizontal asymptote: $y = 1$

Domain: $x \in \mathbb{R}, x \neq -2$

Range: $y \in \mathbb{R}, y \neq 1$

Range : $y \in \mathbb{R}, y \neq 1$